

DAX aggregations

DAX aggregations with measures

What is a measure?

A measure is a **calculation or aggregation** of data in a table or across multiple tables to provide meaningful insights and **summarise** information.

Measure calculations are defined using functions and operators provided by the **DAX language**.

Measures can be **used in** Power BI **visualisations** such as cards, charts, tables, and matrices to **display the calculated result**.

Measures are calculated on the fly (**not stored in the data model**). This makes them **dynamic** which means that they will **adjust to any changes in our data**, without having to manually update the underlying formula.

Importance of measures in Power BI:

- **Performing complex calculations:** Can incorporate complex logic and multiple calculations.
- **Aggregating data:** Commonly used to summarise data and obtain certain insights.
- **Creating metrics:** Allows us to create new metrics that can be used to track performance and monitor KPIs.
- **Dynamic reporting:** Allows us to perform calculations on our data dynamically depending on user interactions.
- **Comparing and analysing data:** Enables us to compare and analyse data from different perspectives.

Data overview

We will be using **two tables**:

Female_labour_force_participation_rate

Contains the **indicator values** for the gender-related indicator **Labour Force Participation Rate, female**, for various African countries and years. The values are expressed as a percentage of the female population aged 15 and older.

Country_ID	Year	Indicator_value
EGY	2022	15.123
EGY	2021	14.636
EGY	2020	14.484
EGY	2019	15.831
EGY	2018	18.383
EGY	2017	22.145
EGY	2016	23.024
EGY	2015	22.779
EGY	2014	23.131
EGY	2013	22.982
EGY	2012	23.173

Country

Contains some **country-related information** about the countries represented in the Female_labour_force_participation_rate table.

Country_ID	Country_name	Major_region	Region	Development	Population size
EGY	Egypt	Africa	Northern Africa	Less developed regions	110990103
ETH	Ethiopia	Africa	Eastern Africa	Less developed regions	123379924
GHA	Ghana	Africa	Western Africa	Less developed regions	33475870
KEN	Kenya	Africa	Eastern Africa	Less developed regions	54027487
MAR	Morocco	Africa	Northern Africa	Less developed regions	37457971
RWA	Rwanda	Africa	Eastern Africa	Less developed regions	13776698
NGA	Nigeria	Africa	Western Africa	Less developed regions	218541212
ZAF	South Africa	Africa	Southern Africa	Less developed regions	59893885

Note: The datasets are linked.

Implicit measures

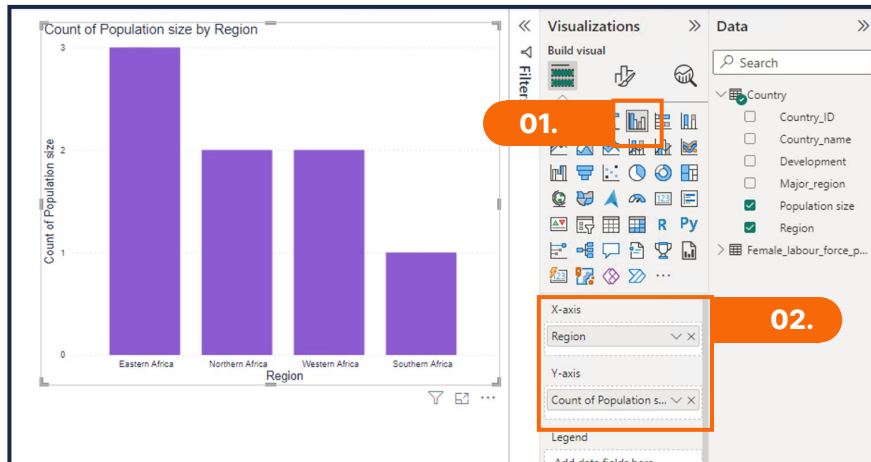
Implicit measures are **created by Power BI automatically** when a **field is used** in a **visual**. Power BI creates and calculates a measure to aggregate the data in that field, then displays the result in the visual created.

A **default aggregation** is **applied** depending on the visualisation type as well as the field data type.

For example:

- 01. Select the clustered column chart** visual type from the **Build visual** pane.
- 02. Drag the Region** field to the **X-axis** well and the **Population_size** column to the **Y-axis** well from the **Country** table.

We see Power BI performed the **Count aggregate function** over the **Population_size** column.



Implicit measures are easy to create but provide us with **limited control** over the calculation.

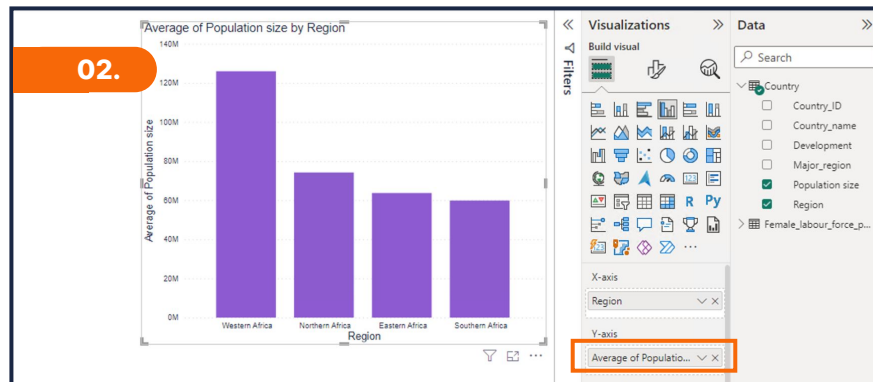
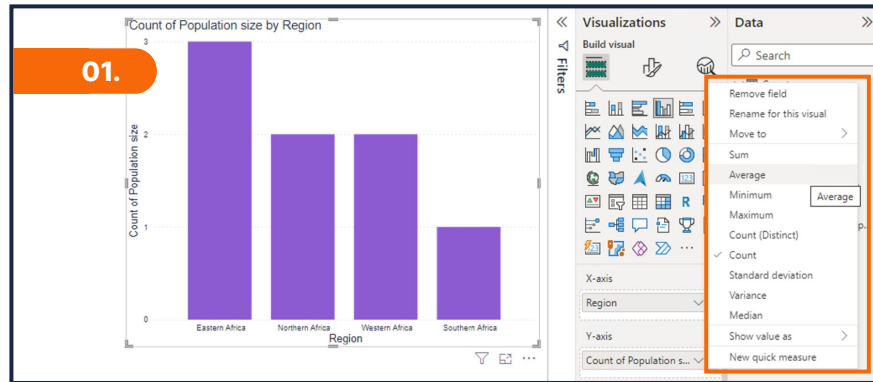
Implicit measures

We can **choose the type of aggregation** to apply to a specific column, or **not to aggregate** at all by choosing from a list of predefined aggregation options.

Steps to change aggregation:

01. Select the visual > Navigate to the **Visualizations** pane and **click** on the **dropdown** next to the field. **Choose** the **desired aggregation** from the menu that appears.
02. The **visual values change** according to the **aggregation selected**. Select the **Average** aggregation.

Our **Population_size** column is now **aggregated by average**.



Explicit measures

Explicit measures are **defined by us** using **DAX expressions**. This gives us **more control** over the aggregation and calculation logic, allowing us to **create more complex and unique calculations and aggregations**.

Advantages of using explicit measures:

- **Flexibility:** They allow us to **control the behaviour** and logic of our **calculations**.
- **Reusability:** Once we define our measure, it is saved and we **can refer to it as many times** as we need in visuals or other measures.
- **Maintainability:** If any calculation logic needs to change, we would **only need to change one** measure, and all referring visuals and measures will update automatically.

Types of explicit measures:

- Quick measures
- Simple measures
- Compound measures

Explicit measures are **more flexible** but they **take more time and effort** since they require us to know how to **use DAX**.

Quick measures

Quick measures are **predefined measures in Power BI** that allow us to quickly create common calculations **without** having to **manually write the DAX query**.

A quick measure **creates and runs** a set of **DAX commands in the background** depending on the input we provide on the Quick measure dialogue box.

Quick measures are **limited** to a **subset of calculations**. We may have to write our own DAX formulas to achieve more flexibility.

We are **shown the DAX formula** used to **implement the measure**, allowing us to modify it to fit our requirements if needed.

Note: Quick measures can be a great way to **solidify our DAX knowledge**.

Quick measures

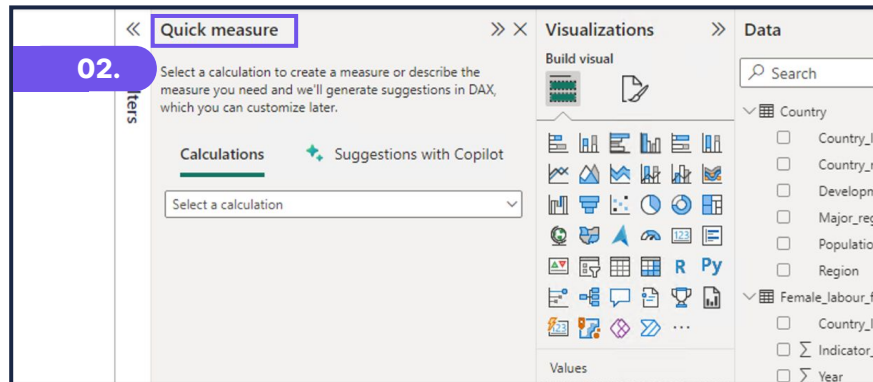
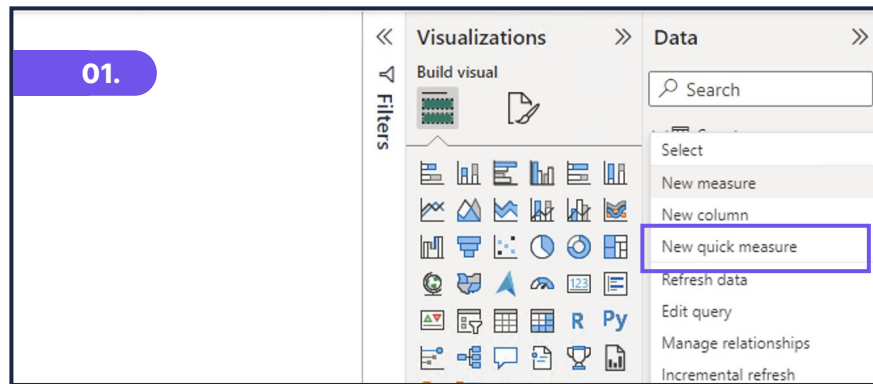
Suppose we wish to create a quick measure that **summarises the average number of countries per region**.

Steps to create a quick measure:

01. **Navigate** to the **table** where we want to add the measure.

Click on the **ellipsis** then select **“New Quick Measure”** from the menu.

02. The **Quick measure window** appears where we choose the calculation we want and the fields to run the calculation against.



Quick measures

03. Click on the **Select a calculation** dropdown.

This produces a list of available **calculation category options** each with multiple calculations under them.

Select the desired **calculation**. We select **“Average per category”**.

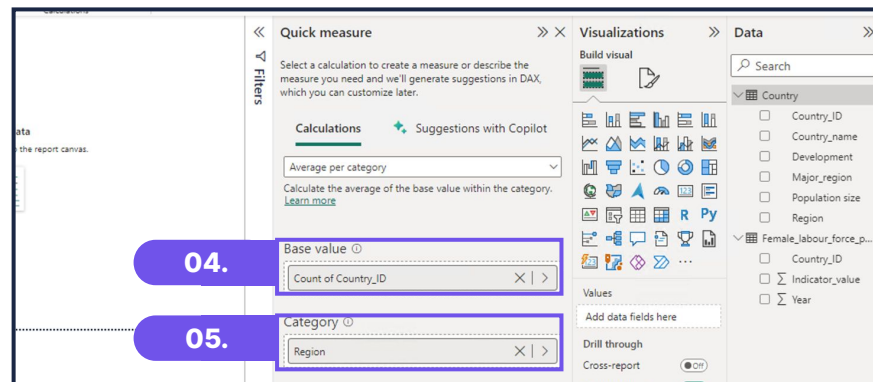
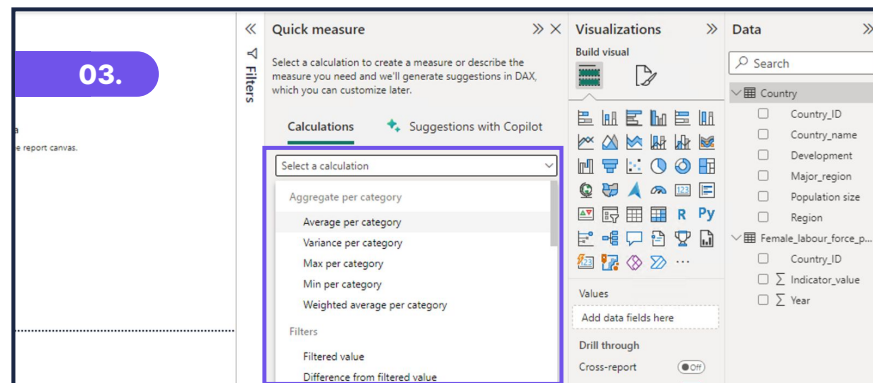
04. **Drag and drop** the **fields** into the **given dialogue box options** which depend on the calculation chosen.

We will **drag** the **Country_ID** field to the **Base value** well.

The **Country_ID** column has been **automatically aggregated** by **Count**.

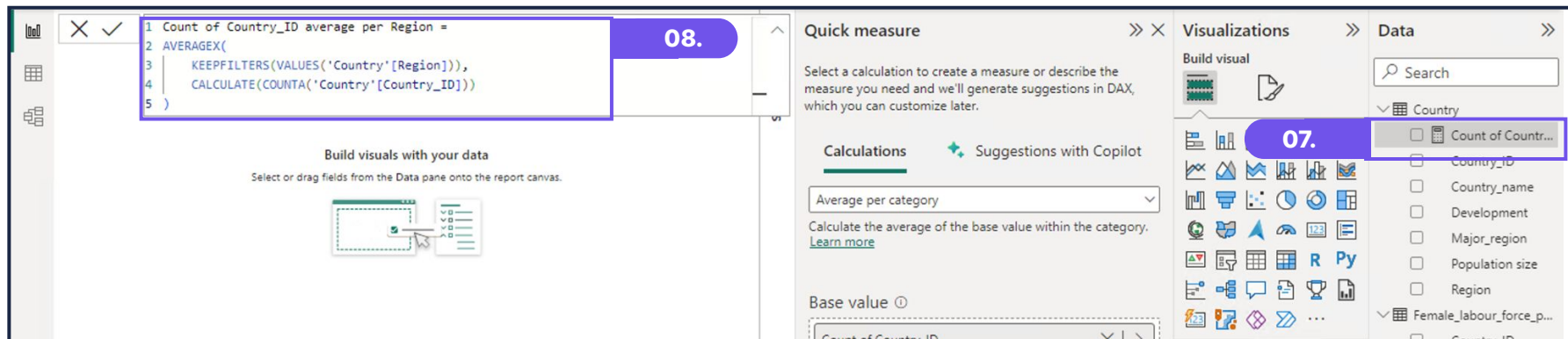
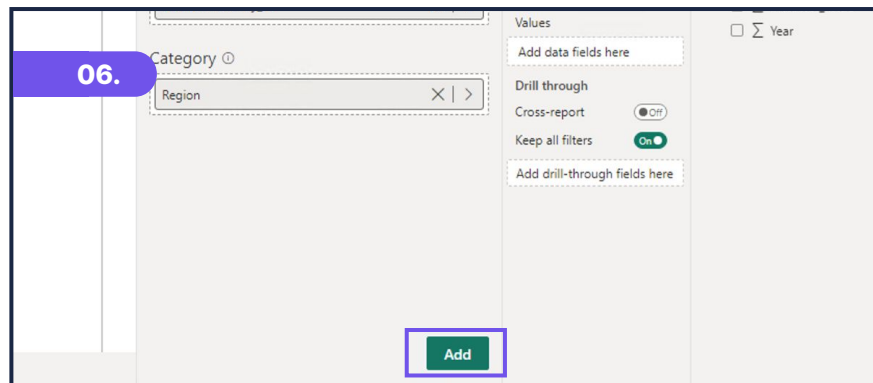
05. Drag the **Region** field to the **Category** well.

Click on the arrow next to the fields in use to **change** the **field** and the **aggregation** to be applied.



Quick measures

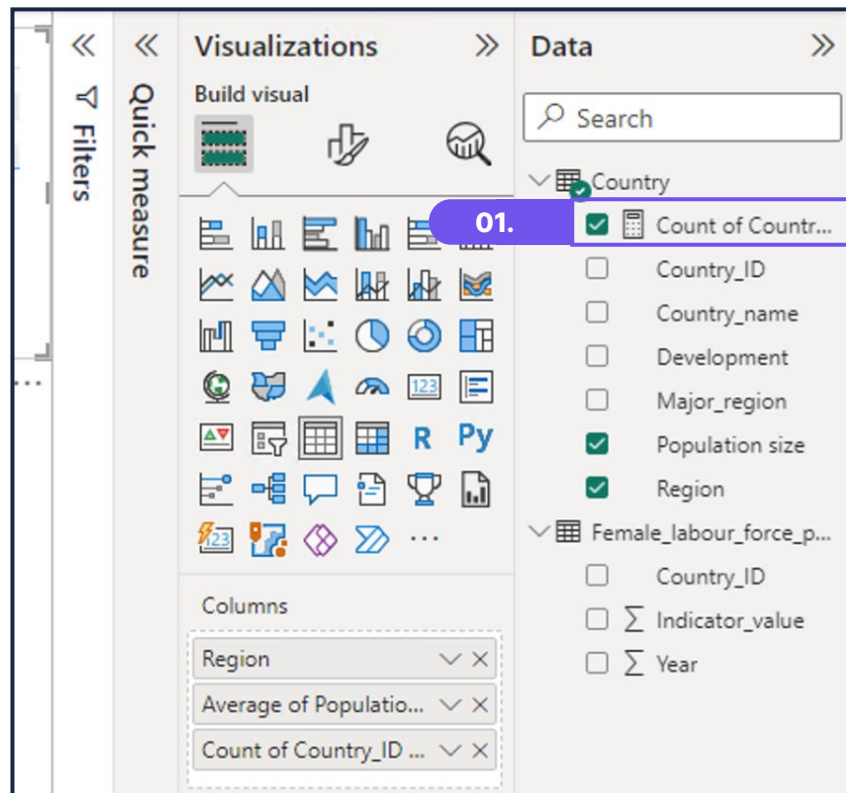
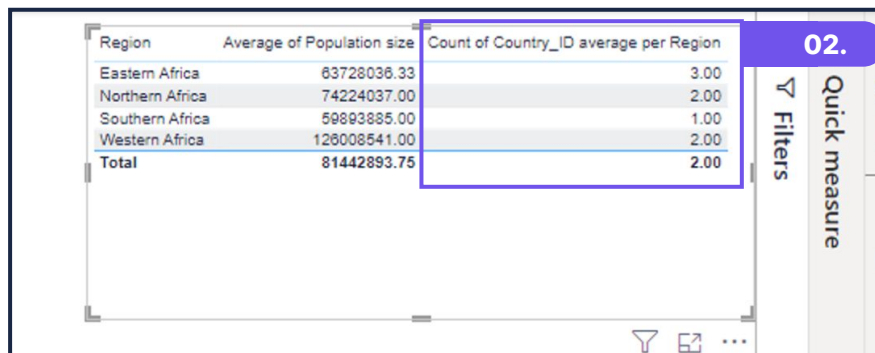
06. Click **Add** after selecting the calculations and fields we want for our quick measure.
07. The **new quick measure** appears in the **Fields** pane.
08. The **underlying DAX formula** appears in the **formula bar**.



Using a measure in a visual

Adding a **measure to a visualisation** in Power BI is similar to how we would normally add a feature. Suppose we already have a visual on our report page:

- 01. Select the visual. Select the measure** we wish to include in the Data pane.
- Its resulting values will be **added** to the selected visual.



Simple measures

Simple measures are **basic calculations** involving a **single expression or aggregation**, similar to implicit measures in construct, without involving multiple measures or complex logic.

Say we want to create a simple measure that calculates and returns the **average female labour force** for the **most recent year** in the dataset, i.e. **2022**.



- The **Average female labour force 2022** measure calculates the average female labour force for the year 2022.
- We use the **AVERAGE** function to calculate the average for the **Indicator_value** column from the **Female_labour_force_participation_rate** table, filtered to the year 2022 using the **CALCULATE** function.

We will use the following **DAX formula**:

```
Average female labour force 2022 =  
    CALCULATE (  
        AVERAGE (  
            'Female_labour_force_participatio  
n_rate'[Indicator_value]),  
            'Female_labour_force_participatio  
n_rate'[Year] = 2022  
        )  
    )
```

Simple measures

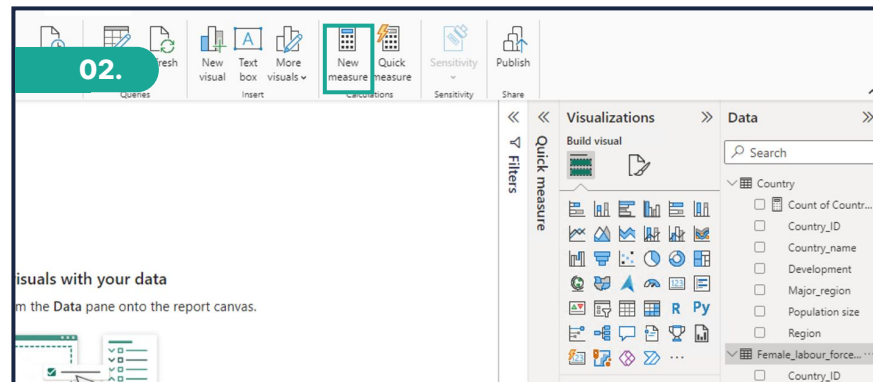
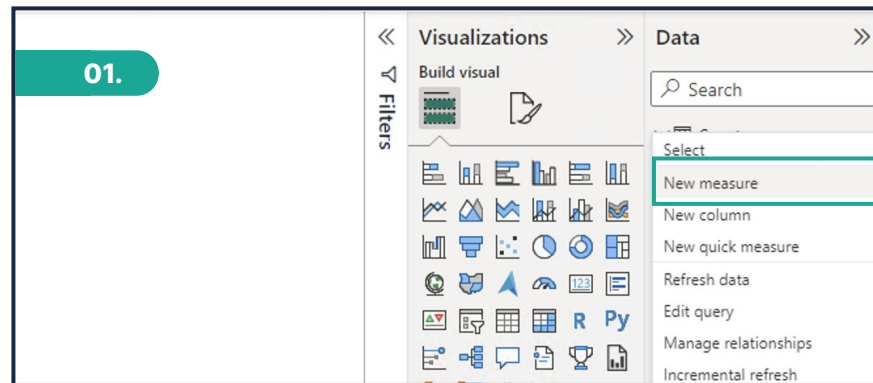
Steps to create a simple measure:

- 01. Navigate to** the `Female_labour_force_participation_rate` **table** containing the required column(s).

Click on the **ellipsis** then select **New measure** from the menu.

or

- 02.** On the **Home** tab > go to the **Calculations** group in the top menu bar and select **New measure**.



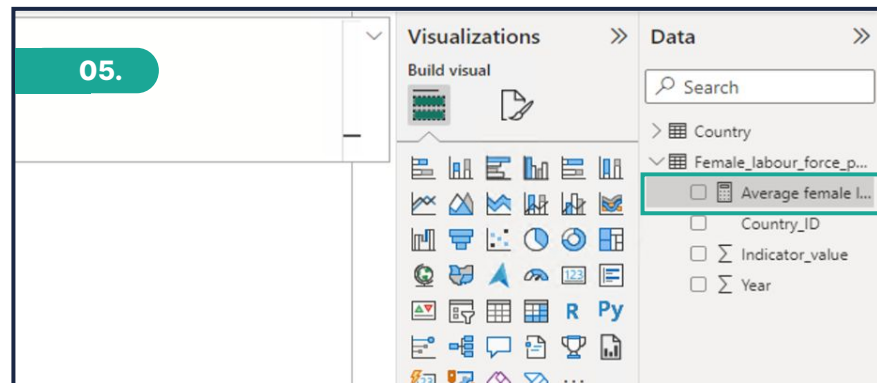
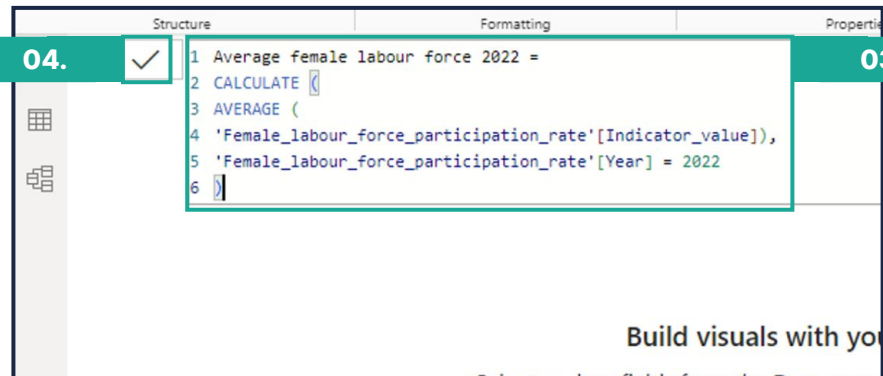
Simple measures

03. The **formula bar** appears at the top.

Specify a **name** for the measure then add a **DAX expression** to perform the desired calculation.

04. Press **Enter** or **click on Commit** (the checkmark icon) for the measure to be published in the table.

05. The **new measure appears** in the **Fields pane** under the table it is based on.



Compound measures

Compound measures are explicit measures that are created by **combining multiple measures or queries**. They are useful when we want to define **more advanced calculations** that can derive new metrics.

Say we want to create a compound measure that calculates the **difference** in the **average female labour force** between the **most recent year**, 2022, and the **previous year**, 2021.



- The **Female labour force year diff** measure calculates the difference between the average female labour force for the most recent year and that of the previous year, 2021.
- We achieve this by subtracting the average female labour force for the year 2021, filtered using the **CALCULATE** function, from our previously created measure **Average female labour force 2022**.

We will use the following **DAX formula**:

```
Female labour force year diff =  
[Average female labour force 2022] -  
CALCULATE (  
    AVERAGE (  
        'Female_labour_force_participation_rate'  
        [Indicator_value]),  
        'Female_labour_force_participation_rate'  
        [Year] = 2021 )
```

Compound measures

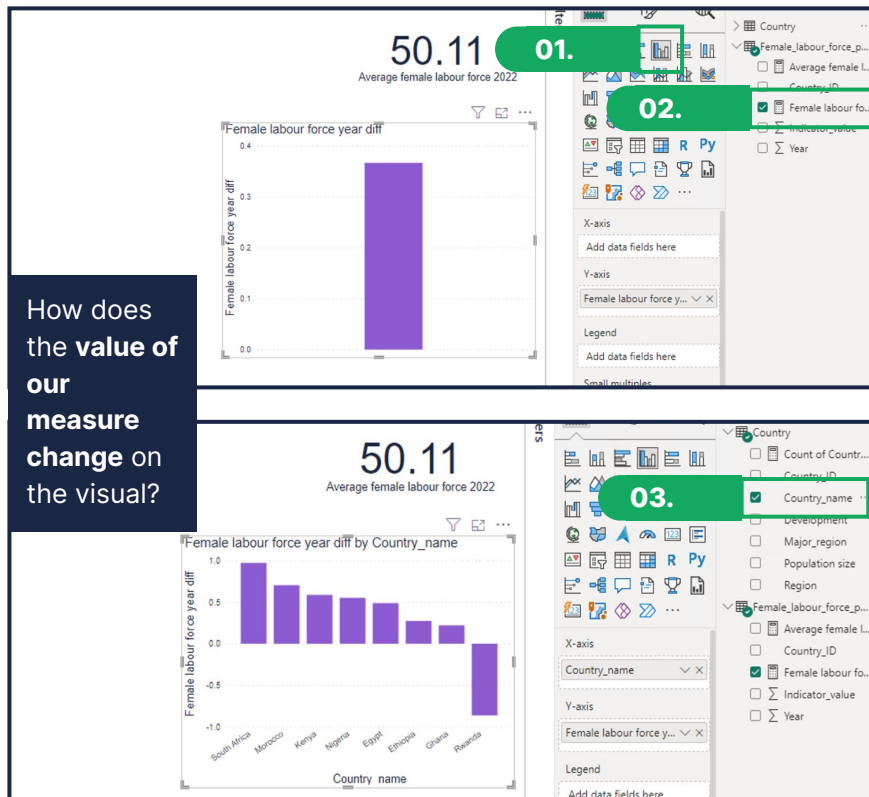
Steps to create a compound measure:

The **steps** for creating a compound measure are **similar to** the ones for a **simple measure**.

The only thing that **changes** is the **DAX formula** we enter.

Add the measure to a visualisation:

- 01.** Select the **Clustered column chart** visual type from the **Build visual** pane.
- 02.** Select the **measure** to add it to the visual.
- 03.** Select the **Country_name** field in the **country table** to add it to the visual.



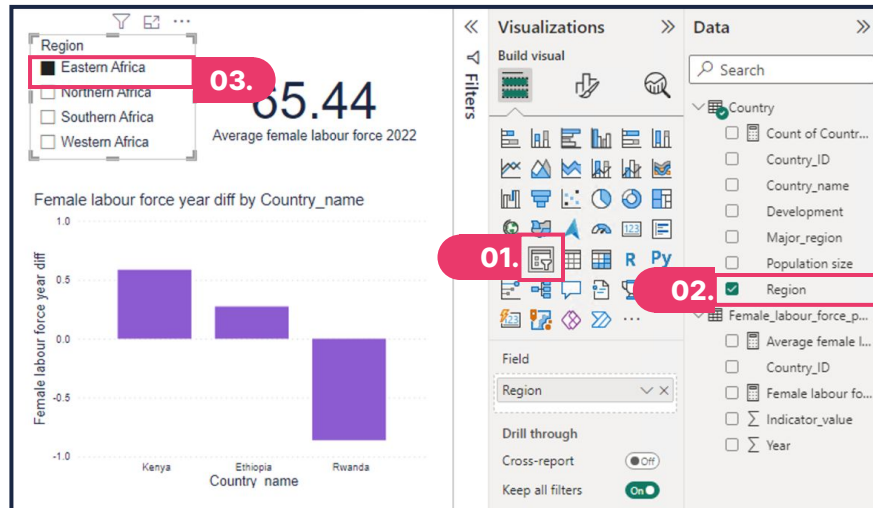
Filter context and its impact on measures

Filter context refers to the **set of filters** that are **applied to the data** at any **given point in time**. It's determined by selections made in slicers, visuals, DAX functions, and interactions within a Power BI report.

Filter context **significantly influences** the behaviour of **measures**, as they **adjust their calculations** based on the **active filters** to provide relevant results.

Add a slicer:

- 01.** Select a **Slicer** from the **Build visual** pane.
- 02.** Add the **Region** field to the slicer.
- 03.** Select the "East Africa" region on the slicer to filter the data on the report accordingly.



How do the visuals that display the results of our measure respond to the filter?